MIS 720 Step 2 Preliminary Analysis — Team B2

From Beats to Hits: Predicting Spotify Song Popularity

Spring 2026

1. Project Recap

This document accompanies our 15-minute Step 2 meeting. It summarizes the preliminary analysis we have completed since submitting the proposal: variable typing, exploratory data analysis (EDA), choice of methods, training/evaluation pipeline, and a first round of modeling results. We have run the full pipeline end-to-end so that the meeting can focus on methodological choices we have made and questions we would like to discuss before Steps 3 and 4.

The project asks whether intrinsic audio features predict whether a Spotify track reaches the dataset's high-popularity threshold (track_popularity > 68). The proposal anticipated examining "whether audio features remain predictive after controlling for genre," so we have structured all of our modeling around two parallel feature sets: an audio-only set and an audio-plus-genre set. We compare them on every model.

2. Data Description and Exploratory Analysis

2.1 Source files, sample size, and variable typing

The Kaggle dataset ships as two CSV files split by the popularity threshold. Both files share an identical 29-column schema. After joining, we identified 295 keys appearing more than once (43 of them appearing in both files), which we resolved by retaining the highest-popularity row per track_id. One row with NA across most audio features was dropped. The resulting working dataset contains 4,494 unique tracks, of which 29.5% (1,325) fall in the High class and 70.5% (3,169) in the Low class.

Variables fall into three groups. We treat the following ten features as continuous predictors: danceability, energy, loudness, valence, acousticness, instrumentalness, speechiness, liveness, tempo, and duration_ms. Five variables are categorical: key (12 levels), mode (2), time_signature (4 effective levels), playlist_genre (35), and playlist_subgenre (84). Eight columns are textual identifiers (track_id, track_name, track_artist, etc.) which we exclude from modeling but retain for traceability. The response variable, high_popularity, is a binary factor derived from track_popularity using the dataset's threshold of 68.

2.2 Distributions of continuous predictors

Histograms of the ten continuous predictors reveal expected patterns: danceability, energy, valence, and tempo are roughly bell-shaped or moderately skewed; acousticness and instrumentalness are heavily right-skewed with most tracks near zero; loudness clusters near 0 dB with a long left tail; and duration_ms is right-skewed but well within reasonable bounds. None of the distributions exhibits pathological behavior that would require transformation prior to modeling, though acousticness and instrumentalness will be standardized for the kernel-based and linear models.

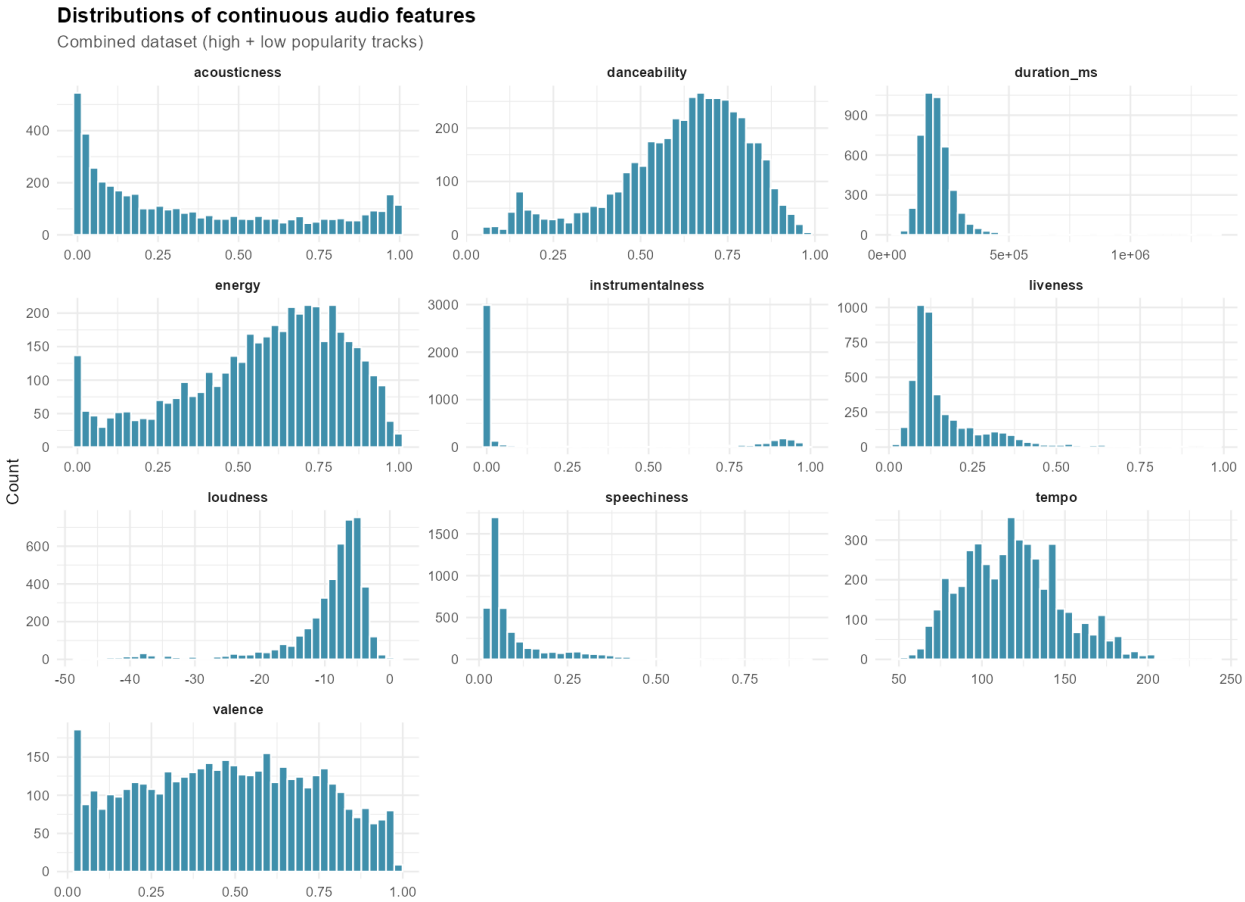

Figure 1. Distributions of the ten continuous audio features.

2.3 Response distribution and threshold

The response track_popularity ranges from 11 to 100 with a mean of 54.8 and a median of 56. Applying the dataset's threshold of 68 yields the 29.5%/70.5% (High/Low) class balance noted above. The proposal's framing as a binary classification problem rather than a regression problem aligns with this pre-partitioned structure.

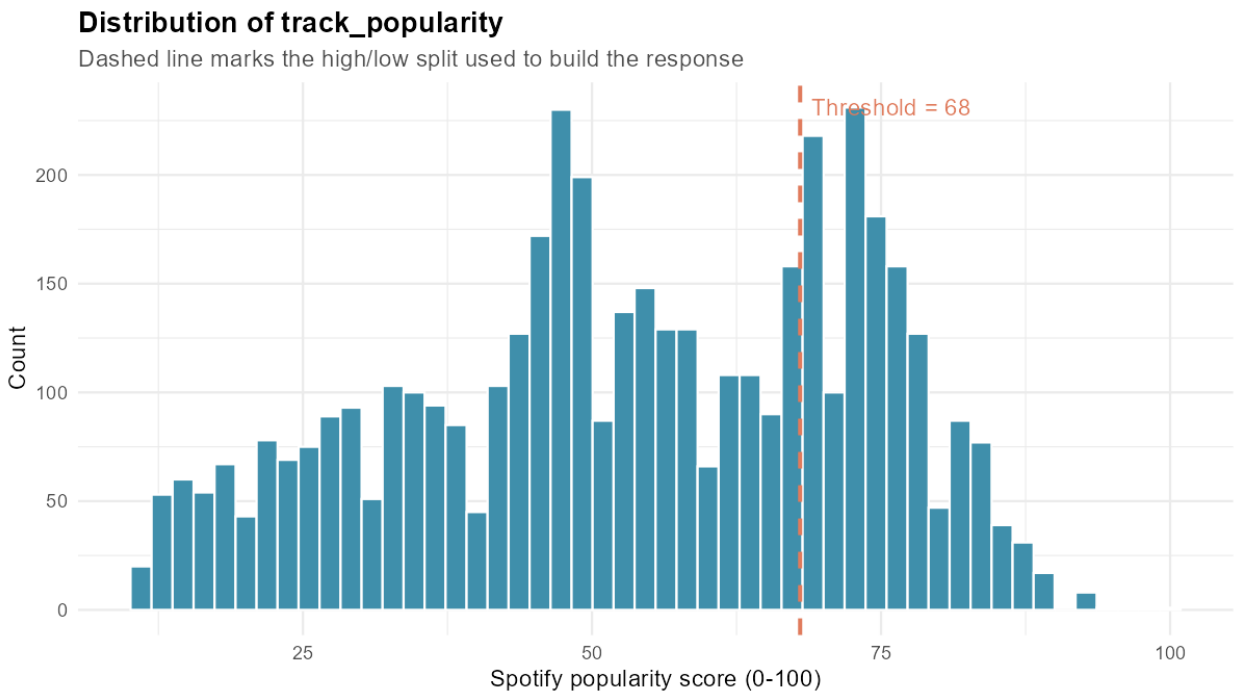

Figure 2. Histogram of track_popularity with the dashed line marking the threshold of 68 used to construct the response.

2.4 Categorical predictors

The five categorical variables vary widely in cardinality. The musical attributes are well-balanced: key has 12 fairly even levels (max 12.2%, min 3.4%), mode is 56% Major / 44% Minor, and time_signature is dominated by 4/4 (88.9%). The playlist labels are far more numerous: playlist_genre has 35 levels topped by electronic, pop, latin, and hip-hop, and playlist_subgenre has 84 levels with no level exceeding 15%. For modeling, we apply step_other(threshold = 0.01) to playlist_genre, which collapses any level appearing in less than 1% of the training set into a single “other” level. We exclude playlist_subgenre from modeling because of its excessive cardinality and overlap with playlist_genre.

2.5 Predictor-response relationships

Because the response is categorical, we examine continuous-predictor relationships using side-by-side box plots and categorical-predictor relationships using stacked proportion bars. The continuous box plots show the largest separations on instrumentalness (High tracks much lower), acousticness (High lower), loudness (High higher), and energy (High higher). These four directions match the proposal’s a-priori hypotheses. Effects on valence, danceability, tempo, and duration_ms are visible but smaller, and speechiness, liveness, and the audio-categorical features (key, mode, time_signature) show essentially no class separation.

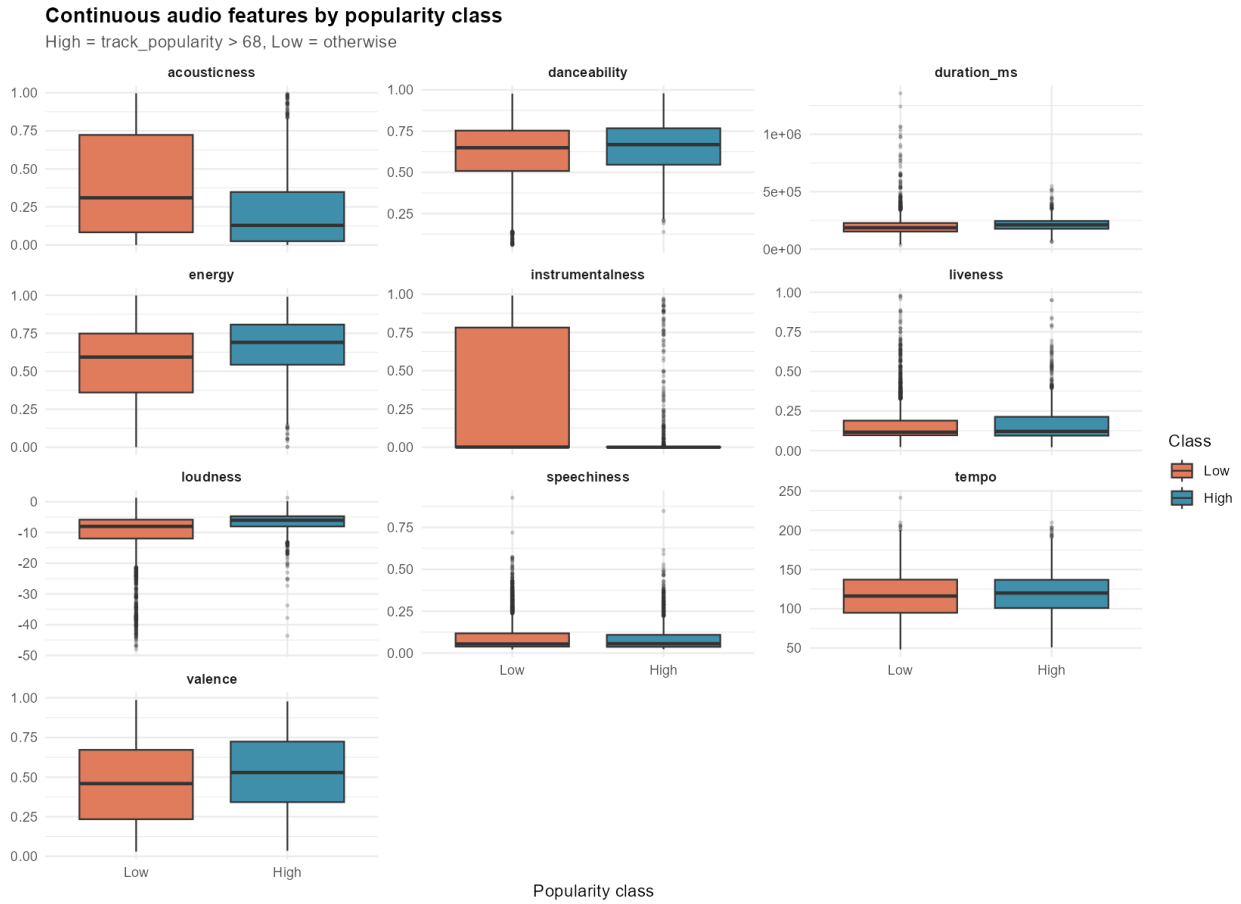

Figure 3. Continuous audio features by popularity class. Boxes summarize within-class quartiles; the four most class-separating features are instrumentalness, acousticness, loudness, and energy.

For the categorical predictors, the stacked-proportion plot of playlist_genre shows the largest signal of any single predictor in the dataset: the High-popularity share varies from over 50% (pop, hip-hop, R&B, rock, gaming) down to nearly zero (ambient, world, jazz, lofi, several international genres). This is the strongest single feature we observe and motivates the audio-vs-audio-plus-genre comparison that anchors our modeling.

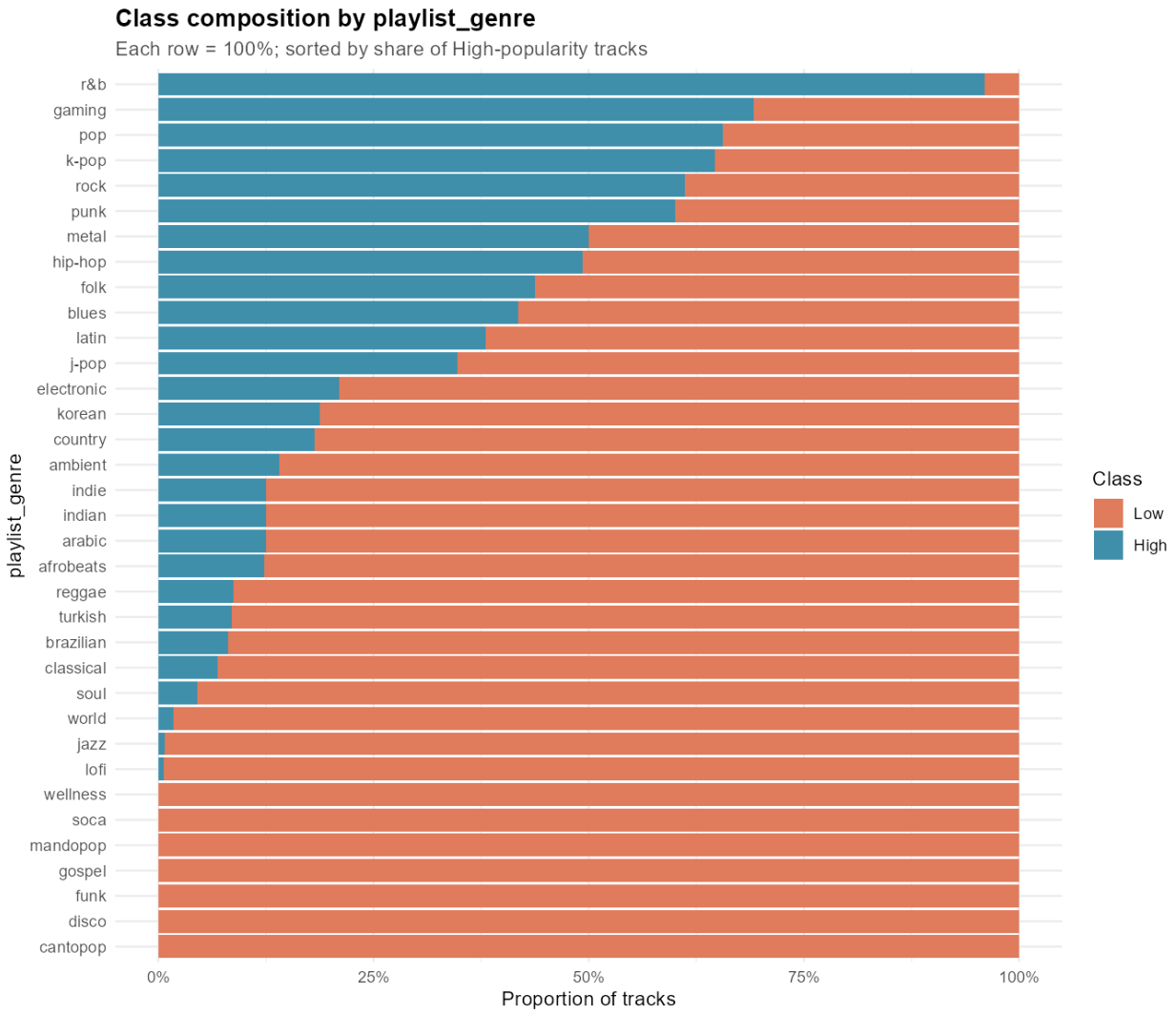

Figure 4. Class composition by playlist_genre, sorted by share of High-popularity tracks. Each row sums to 100%.

2.6 Multicollinearity among continuous predictors

Pairwise Pearson correlations confirm the proposal's expectations. The strongest absolute correlations involve energy and loudness ($r = 0.80$), energy and acousticness ($r = -0.75$), loudness and acousticness ($r = -0.65$), and loudness and instrumentalness ($r = -0.64$). All other pairs are below $|r| = 0.6$. We do not remove any feature on collinearity grounds, but we note that logistic-regression coefficients on energy, loudness, acousticness, and instrumentalness should be interpreted jointly rather than individually.

3. Modeling Approach

3.1 Methods

We selected three of the methods enumerated in the course outline. Logistic regression is the required linear baseline given a categorical response, and provides interpretable coefficients in standardized units. Random forest is our nonlinear ensemble baseline; it handles factor predictors natively and yields direct feature-importance rankings. SVM with an RBF kernel is our kernel-based discriminative comparator. Decision trees and neural networks are reserved as possible additions for the final report if there is time.

Method	Engine	Role
Logistic Regression	stats::glm	Required linear baseline; interpretable standardized coefficients.
Random Forest	ranger (500 trees)	Nonlinear ensemble; native factor handling; tuned mtry and min_n on a 20-point grid.
SVM (RBF kernel)	kernlab	Kernel discriminative; tuned cost and rbf_sigma on a 20-point space-filling grid over [2^-2, 2^8] x [10^-4, 10^-1].

3.2 Feature sets

We train every model on two parallel feature sets. The audio set uses the ten continuous audio features plus key, mode, and time_signature. The audio-plus-genre set adds playlist_genre. Comparing the two lets us evaluate how much of any model's predictive power is attributable to audio characteristics alone versus the playlist-assigned genre. This produces six fitted workflows in total (three models × two feature sets).

3.3 Pipeline

The dataset is split 80/20 into a training set (3,595 rows) and held-out test set (899 rows), stratified on the response. Within the training set, we use 5-fold stratified cross-validation for hyperparameter tuning. Logistic regression has no tunable hyperparameters and is evaluated directly on the resamples. The two tunable models use ROC AUC as the tuning objective because it is threshold-independent and does not collapse under the moderate class imbalance. Tuning metrics are limited to threshold-independent metrics (ROC AUC, accuracy, Brier score) to avoid undefined precision/F1 values for hyperparameter combinations that predict no positives at the 0.5 threshold. After tuning, each workflow is finalized at its CV-best hyperparameters, refit on the full training set, and evaluated once on the held-out test set with the full metric set.

Preprocessing is applied via tidymodels recipes. For logistic regression and SVM the recipe imputes any remaining missing values (median for numerics, mode for nominals), collapses rare playlist_genre levels with step_other(threshold = 0.01), one-hot encodes categorical predictors, and z-standardizes all numeric predictors. The random forest uses the same imputation and rare-level collapsing but no encoding or scaling, since ranger handles factors natively and trees are scale-invariant.

4. Preliminary Results

4.1 Test-set metrics

Test-set metrics for all six workflows are summarized in Table 2 at the default decision threshold of 0.5. Three patterns are consistent across all three model families. First, adding playlist_genre yields a sizable lift on every metric, with ROC AUC improving by 0.07–0.17 and accuracy by approximately eight percentage points. Second, random forest on audio-plus-genre is the best performer across nearly every metric (ROC AUC 0.872, F1 0.613). Third, sensitivity (recall on the High class) is strikingly low for every audio- only model at this threshold (0.09–0.20), and only moderate for audio-plus-genre (0.48–0.56).

The low sensitivity is not a property of the models themselves but of the interaction between a 30/70 class prior and a fixed 0.5 cutoff: the predicted-probability distributions for High tracks are pushed below 0.5 for many true Highs, even though the underlying ranking (captured by ROC AUC and Brier score, which are threshold-independent) remains useful. Section 4.5 reports test-set metrics at a CV-tuned threshold per workflow, where sensitivity rises substantially. The default-0.5 numbers are presented first because they are the conventional reporting baseline; the tuned-threshold numbers in §4.5 are what we would emphasize for any deployment scenario.

Model	Feature set	Acc.	AUC	Sens.	Spec.	Prec.	F1	Brier
Logistic Regression	audio	0.703	0.731	0.094	0.957	0.481	0.158	0.178
Logistic Regression	audio+genre	0.784	0.849	0.562	0.877	0.656	0.606	0.142
Random Forest	audio	0.724	0.760	0.200	0.943	0.596	0.299	0.172
Random Forest	audio+genre	0.803	0.872	0.528	0.918	0.729	0.613	0.132
SVM (RBF)	audio	0.685	0.674	0.151	0.909	0.408	0.220	0.196
SVM (RBF)	audio+genre	0.778	0.842	0.475	0.904	0.674	0.558	0.162

Table 2. Test-set metrics at the default threshold of 0.5. Sens./Spec./Prec./F1 are computed with High as the positive class.

4.2 Discrimination and ranking quality

ROC curves for all six workflows confirm the metric table. On audio-only inputs, all three models produce moderate discrimination (AUC 0.67–0.76); on audio-plus-genre, all three reach AUC ≥ 0.84. The random forest curve dominates the others on audio-plus- genre across most operating points, while logistic regression and SVM are nearly indistinguishable on the same feature set.

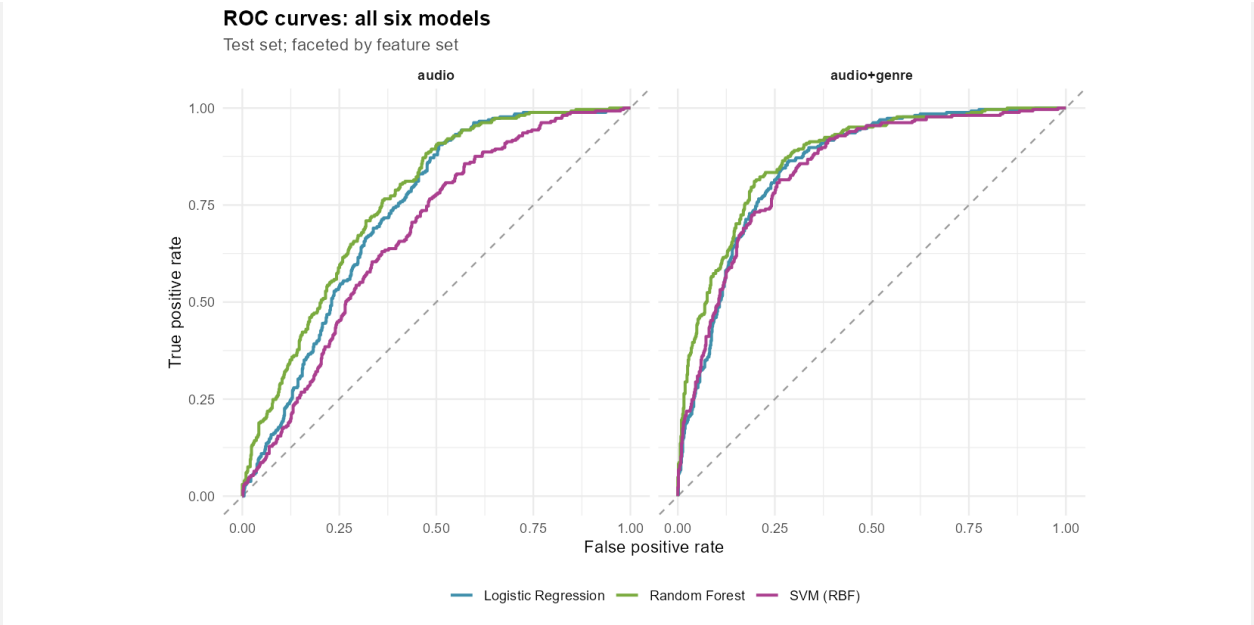

Figure 5. ROC curves on the held-out test set, faceted by feature set and colored by model.

4.3 Cross-model metric comparison

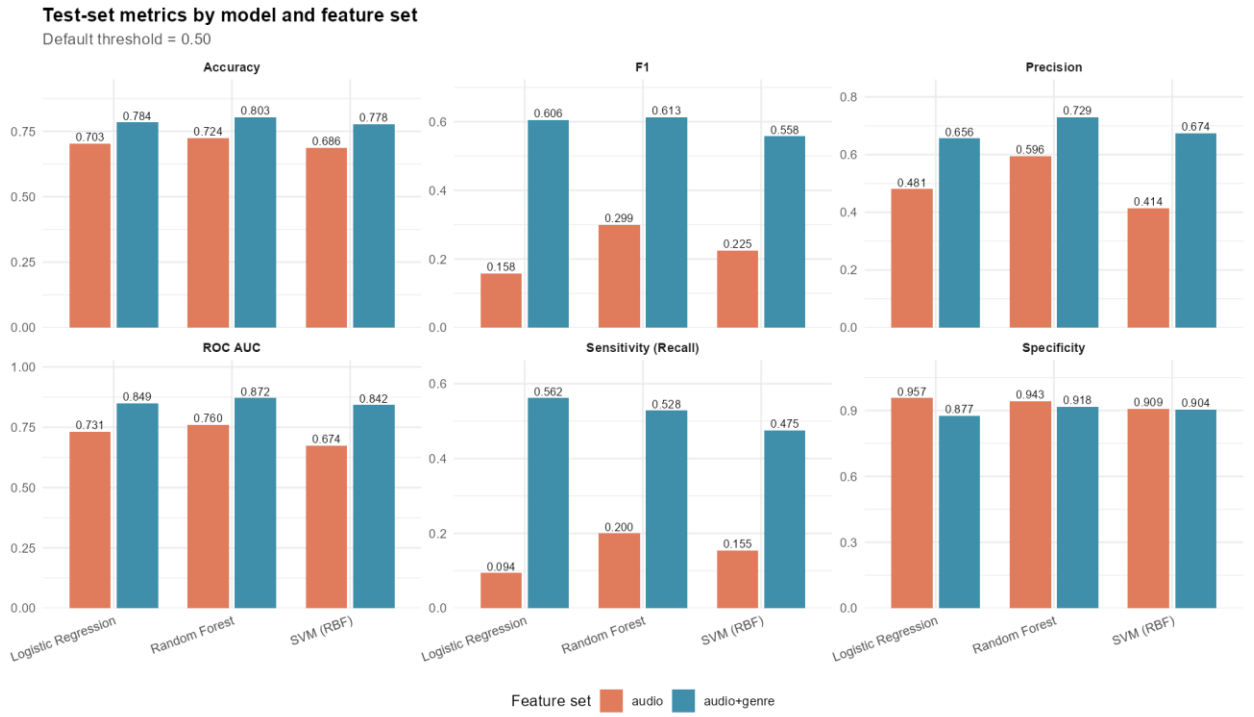

Figure 6. Test-set metrics by model and feature set. The audio-vs-audio+genre gap is consistent across all metrics.

4.4 Feature attribution

Permutation importance measured on the test set as the drop in ROC AUC when each feature is shuffled gives a uniform attribution scale across all three model families. The pattern is consistent: when playlist_genre is included, it dominates by a wide margin, accounting for

approximately 0.18–0.24 drops in ROC AUC across the three models. Among the audio features, instrumentalness and loudness are the next two most important everywhere; danceability and duration_ms emerge for some models; key, mode, time_signature, liveness, and tempo contribute negligibly. When genre is removed, the relative ordering among audio features is preserved but the absolute importance of the leading features (instrumentalness, loudness) increases, consistent with the model leaning more heavily on what audio signal it has.

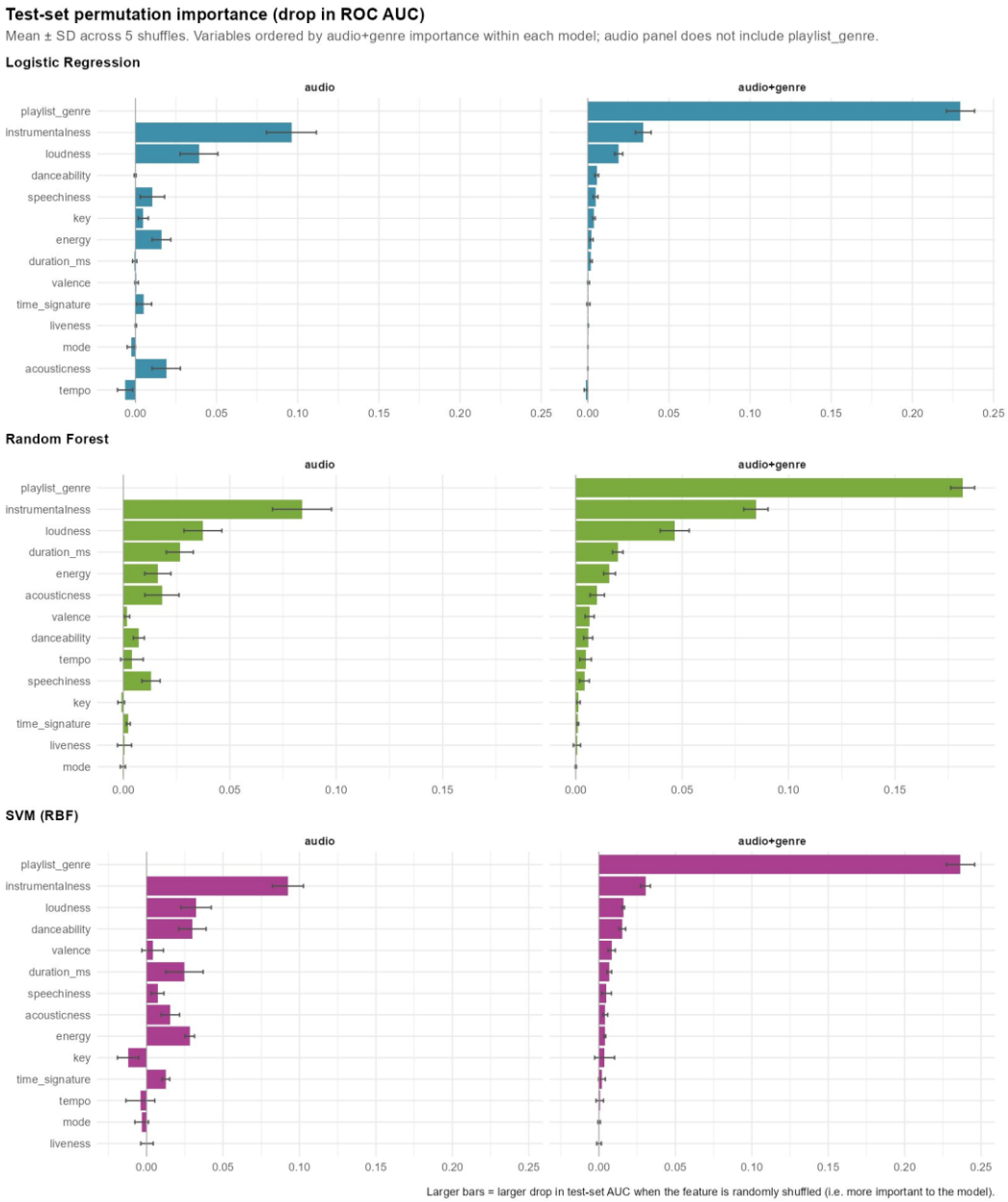

Figure 7. Permutation importance on the test set for all six workflows, expressed as the mean drop in ROC AUC over five shuffles per feature.

4.5 Threshold tuning

Because the default 0.5 threshold suppressed sensitivity (§4.1), we tuned the decision threshold separately for each workflow. The threshold is selected on the cross-validation out-of-fold

predictions to maximize F1 on the High class, then applied once to the held-out test set. This keeps the test set genuinely held out: no test data informs the threshold choice.

Selected thresholds range from 0.19 to 0.33 across the six workflows, all well below 0.5 — consistent with the 30/70 class prior pulling the F1-optimal cut leftward. The F1 sweep in Figure 8 shows that on audio-plus-genre, the F1 curves for logistic regression and random forest form broad plateaus (F1 ≈ 0.65 across thresholds 0.25–0.55), suggesting those models are robust to small threshold misspecification. The audio-only curves and the SVM audio-plus-genre curve peak more sharply, the latter at the lowest threshold of any workflow (0.19), which is consistent with kernlab's known difficulty producing well-calibrated probabilities.

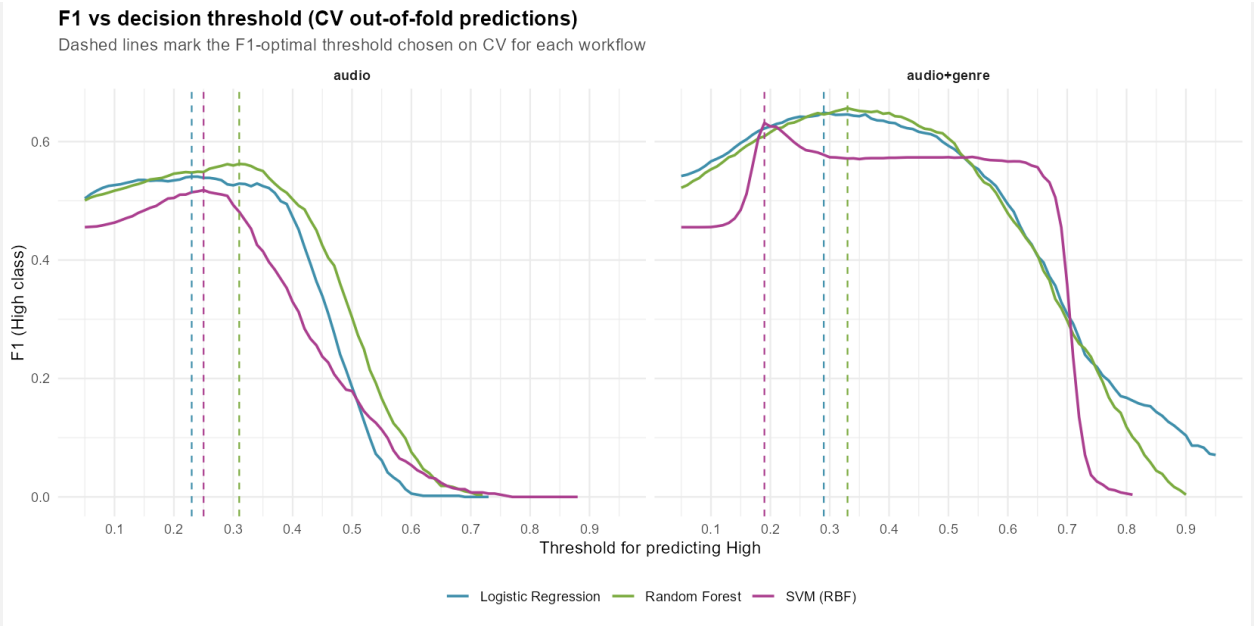

Figure 8. F1 as a function of the decision threshold, computed on cross-validation out-of-fold predictions. Dashed verticals mark the F1-optimal threshold chosen per workflow.

Test-set metrics at the CV-tuned thresholds are shown in Table 3 alongside the default-0.5 numbers from Table 2. The lift in sensitivity is substantial across the board: the audio-only logistic regression rises from 0.094 to 0.936, the audio-plus-genre random forest from 0.528 to 0.834, and every other workflow follows the same pattern. Specificity falls correspondingly, but only to the 0.44–0.76 range — still above chance everywhere and well above the 0.5 floor for the audio-plus-genre models. F1 improves on every workflow, by 0.08–0.41 absolute. Accuracy decreases slightly because the prior-weighted accuracy metric over-rewards correctly predicting the majority Low class, which the default threshold did frequently and the tuned threshold does less.

Model	Feature set	Scheme	Thr.	Sens.	Spec.	Prec.	F1	Accuracy
Logistic Reg.	audio	0.5	0.50	0.094	0.957	0.481	0.158	0.703
Logistic Reg.	audio	tuned	0.23	0.936	0.440	0.411	0.571	0.586
Logistic Reg.	audio+genre	0.5	0.50	0.562	0.877	0.656	0.606	0.784

Model	Feature set	Scheme	Thr.	Sens.	Spec.	Prec.	F1	Accuracy
Logistic Reg.	audio+genre	tuned	0.29	0.826	0.740	0.570	0.675	0.765
Random Forest	audio	0.5	0.50	0.200	0.943	0.596	0.299	0.724
Random Forest	audio	tuned	0.31	0.804	0.591	0.451	0.578	0.654
Random Forest	audio+genre	0.5	0.50	0.528	0.918	0.729	0.613	0.803
Random Forest	audio+genre	tuned	0.33	0.834	0.756	0.588	0.690	0.779
SVM (RBF)	audio	0.5	0.50	0.155	0.909	0.414	0.225	0.686
SVM (RBF)	audio	tuned	0.25	0.755	0.521	0.397	0.520	0.590
SVM (RBF)	audio+genre	0.5	0.50	0.475	0.904	0.674	0.558	0.778
SVM (RBF)	audio+genre	tuned	0.19	0.743	0.763	0.568	0.644	0.758

Table 3. Test-set metrics at the default threshold of 0.5 vs the CV-tuned F1-optimal threshold per workflow. Sensitivity rises substantially across all six workflows; specificity decreases correspondingly but remains above chance.

At the CV-tuned thresholds, the random forest on audio-plus-genre remains the best performer (F1 0.690), but logistic regression on the same feature set is now competitive (F1 0.675). On audio-only inputs, the three models cluster within a 0.058 F1 band (0.520–0.578). Threshold tuning narrows the model-to-model gaps because each model now operates at its own appropriate decision point rather than being held to a uniform 0.5 cut.

5. Concerns and Questions for Discussion

We would like to use the meeting to discuss the following items, which involve methodological choices we have made tentatively and would like to validate before producing the final report.

5.1 Reporting threshold for the final report

Section 4.5 demonstrated that the choice of decision threshold materially affects how each model's performance is summarized. The default 0.5 threshold favors specificity at the expense of sensitivity (a property of the 30/70 class prior, not of the models), while the CV-tuned thresholds (0.19–0.33) produce balanced operating points where the models actually identify a useful share of the High-popularity tracks. Both threshold choices draw on the same underlying probabilistic predictions; they are different summarizations of the same model rather than different models.

We have a clear preference: report the CV-tuned numbers as the headline performance, with the default-0.5 numbers retained for reference. F1 is meaningfully better at the tuned thresholds for every workflow, and the tuned operating points are the ones a deployment would actually use. However, the conventional reporting baseline is 0.5, and we would like to confirm with the instructor whether reporting the tuned threshold as the headline (with explicit disclosure of the tuning procedure) is the preferred presentation, or whether the final report should retain the default-0.5 numbers as headline and treat threshold tuning as a sensitivity analysis.

We did not pursue alternative class-imbalance remedies (class weights or training-set resampling such as SMOTE). The proposal explicitly avoided resampling, and we found threshold tuning sufficient to address the visible imbalance issue. We can revisit this if the instructor recommends a complementary approach.

5.2 Genre as predictor and possible information leakage

The dataset's playlist_genre is derived from the playlist a track was sourced from rather than from the track itself. Tracks appearing on multiple playlists were deduplicated by keeping the highest-popularity row, which means each track ends up tagged with the genre of its most-popular containing playlist. This raises a question of whether playlist_genre is acting as a true predictor of audio-derived popularity or as a partial proxy for curatorial signal (a track on a popular pop playlist is more likely to be a hit in part because it is on a popular playlist). The 0.07–0.17 lift in AUC and dominance in permutation importance are large enough that we want to flag this. Our current plan is to report both feature sets and discuss the framing question explicitly in the final report; we would welcome the instructor's view.

5.3 SVM (RBF) underperformance

SVM with the RBF kernel consistently lags both other models, by a small margin on audio-plus-genre and a larger one on audio-only (test AUC 0.674). Our hyperparameter search was non-trivial: an initial run hit a degenerate corner of the default rbf_sigma range and required us to constrain the search space. Even after the fix, SVM does not match either alternative on this dataset. We see two interpretations: (i) the data's decision boundary is well-approximated by the linear and tree-ensemble models, leaving the kernel SVM with little to add, or (ii) further tuning (e.g., a wider grid or Bayesian optimization) could close the gap. We will keep SVM in the final report for the comparison but will note the underperformance honestly.

5.4 Whether to add a fourth model

The course outline lists six possible methods. We chose three (logistic regression, random forest, SVM) and the project guidelines require at least three. We considered adding a decision tree as a more interpretable counterpart to random forest, or a small feed-forward neural network. We are open to either; our default is to keep the model count at three for the meeting and add a fourth only if the instructor recommends it.

5.5 Final-report structure

The final report is required to be 10–15 pages. Our current plan is to allocate roughly two pages to dataset and EDA, two to methodology, four to results (including model comparison and feature attribution), and three to interpretation, limitations, and the audio-vs-genre framing discussion. We would like to confirm this allocation matches the instructor's expectations before drafting.

6. Generative AI Disclosure

In accordance with the MIS 720 project guidelines, we disclose that generative AI assistance (Anthropic's Claude) was used during this Step 2 phase to scaffold and review the R scripts that produced the EDA figures and modeling results, to assist with debugging (notably an SVM hyperparameter-range issue and a yardstick event-level issue), and to help draft and polish the prose of this document. All substantive decisions — dataset scope, response-variable framing, the audio-vs-audio-plus-genre comparison, the choice of methods and metrics, and every figure and table interpretation — were made by the team. All AI-assisted code was executed locally and inspected before use, and all AI-generated text was reviewed and edited before inclusion.